Module 09 — Spring Data JPA and Validation

Phase: Backend (persist JSON API)
Previous module: Spring Web REST
Next module: REST, DTOs, and Postman
Lab kit: 15_Lab_Assets/18_jpa_product_crud/

Module Outcome:

- Persist a Product with `@Entity` + `JpaRepository` instead of a list in RAM
- Wire MySQL in `application.yml` without committing the password
- List / get / create against the real table — then peek at `@Valid`

Teach order today: in-memory pain → JDBC vs JPA → `@Entity` Product → `JpaRepository` → pom starters + driver → `application.yml` + env + Hikari name + lab `ddl-auto` → REST list / get / create against the DB → lite `@Valid` peek.

Prerequisites

- **Campus Hello REST** from Spring Web REST — `@RestController`, GET list / GET by id / POST, 200 / 201 / 404, in-memory `List`. JDK 17 + Maven 3.9+ + Spring Boot 3.x (Tested Toolchain).
- **MySQL 8+** and **JDBC CRUD** — `MYSQL/` `training_db`, FS bridge CampusCart `products` idea, JDBC CRUD secrets habit (env / `.env.example`).

Definition / Purpose

Definition: JPA (Jakarta Persistence API) is the **Java standard for mapping classes to tables**. An `@Entity` is a class whose fields are columns. `@Id` marks the primary key. **Hibernate** (inside `starter-data-jpa`) implements JPA and writes the SQL.

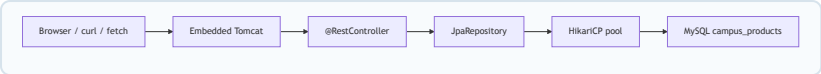
Definition: Spring Data JPA gives you a `JpaRepository`. You declare an interface; Boot supplies `save`, `findAll`, `findById`, `deleteById`. You do **not** write `SELECT *` by hand for those four calls.

Purpose: Priya's CampusCart catalog cannot live in an `ArrayList`. Restart wiped Idli. JDBC already saves rows — but every INSERT is a string of SQL. JPA lets the same `Product` object that Jackson JSON uses **become a row**. Interviews still ask you to contrast JDBC and JPA. **Later —** DTOs so the JSON shape is not the table shape.

🏦 Bank 📦 Order 💻 Laptop 👤 Employee 📱 UPI

Where used?
- CampusCart / **Swiggy** — catalog JSON that **survives restart**
- Canteen / mess — snacks in MySQL, not in the JVM
- **HDFC Bank** / **PhonePe** — account rows behind a JSON list (**no money movement here**)
- **Apollo Hospitals** — medicine / token catalog
- **Employee Management Portal** — employee master (**Later —** relations)

Visual — REST to MySQL:



Key takeaways:

- **JDBC** = you write SQL. **JPA** = you map a class; Hibernate writes SQL
- Core path: **`@Entity` + `JpaRepository` + MySQL`**, same GET/POST as module 02
- Secrets stay in **env**; **Hikari** is auto-configured (name it, do not tune it)
- **`ddl-auto=update` is lab-only**

Motivation — Feel the pain of restart

Ravi POSTed Pani puri. The canteen screen showed it. He restarted `mvn spring-boot:run`. The snack was gone. Workbench still had yesterday's MYSQL homework — the Boot app never used it.

```
public class RestartWipesList {
    public static void main(String[] args) {
        System.out.println("POST /canteen/items -> Pani puri in ArrayList");
        System.out.println("Restart Tomcat -> seed snacks only");
        System.out.println("Need MySQL rows that survive Ctrl+C");
    }
}
```

Expected Output:

```
POST /canteen/items -> Pani puri in ArrayList
Restart Tomcat -> seed snacks only
Need MySQL rows that survive Ctrl+C
```

Remember: Module 02 practised HTTP. This module practises **persistence**. Same mappings; different store.

Interview: "Why not keep the `ArrayList`?" — Demos and tests. Production catalogs are shared and durable. That is a database.

Important Concepts

1. JDBC vs JPA — same MySQL, different Java

Definition: JDBC is the **low-level API**: `Connection`, `PreparedStatement`, `ResultSet`. You already listed library books that way. **JPA** is an **ORM** (object–relational mapping): a `Product` instance ↔ a row.

Purpose: Without the contrast, juniors think JPA "replaces MySQL." It does not. Hibernate opens a JDBC connection from **HikariCP** and still sends SQL. You skip typing `INSERT INTO` for the happy path.

	JDBC (already taught)	JPA (this module)
You write	SQL + `?`	Entity + repository method

	JDBC (already taught)	JPA (this module)
Driver	Connector/J on `-cp`	`mysql-connector-j` in `pom.xml`
Pool	None (one `Connection` per CLI)	**HikariCP** auto-configured
Config	env / `.env`	`application.yml` + env
Restart	Rows stay in MySQL	Rows stay in MySQL

Point at tables you already know — do **not** redo DDL textbooks:

- `MYSQL/` — `CREATE TABLE`, PK, DML
- FS bridge — CampusCart `products` (`product_id`, `name`, `price`)
- Lab table here is **`campus_products`** so we do **not** smash MYSQL homework `products`

```
public class JdbcVsJpaPrint {
    public static void main(String[] args) {
        System.out.println("JDBC: PreparedStatement INSERT INTO books ...");
        System.out.println("JPA:  products.save(new Product(\"Idli\", 40))");
        System.out.println("Both: MySQL 8 + Connector/J + no password in git");
    }
}
```

Expected Output:

```
JDBC: PreparedStatement INSERT INTO books ...
JPA:  products.save(new Product("Idli", 40))
Both: MySQL 8 + Connector/J + no password in git
```

Note:
- Wrong: “JPA means we can forget SQL.” Interviews still ask JOIN and `PreparedStatement`.
- Wrong: mixing `javax.persistence` on Boot 3 — use **`jakarta.persistence`**.
- Prefer: JDBC for “I can write the SQL”; JPA for “the catalog API.”

Common Error: Opening `DriverManager.getConnection` inside a `@RestController` “because JDBC week did that.” Boot already has a **DataSource**. Use the repository.

Remember: JPA sits **on** JDBC. It is not a second database.

Interview: ORM? Object–relational mapping — Java object to table row.

Print who writes SQL this week versus JDBC week: you wrote INSERT/SELECT then; Hibernate writes it now, and you still read it with `show-sql`.

```
public class WhoWritesSql {
    public static void main(String[] args) {
        System.out.println("JDBC week: you wrote INSERT / SELECT");
        System.out.println("JPA week: Hibernate writes INSERT / SELECT");
        System.out.println("You still read the SQL in the log (show-sql)");
    }
}
```

Expected Output:

```
JDBC week: you wrote INSERT / SELECT
JPA week: Hibernate writes INSERT / SELECT
You still read the SQL in the log (show-sql)
```

Print the same secrets habit as JDBC: `MYSQL_PASSWORD` from env, YAML `${MYSQL_PASSWORD}`, `.env.example` in git, `.env` not in git.

```
public class SameSecretsHabit {
    public static void main(String[] args) {
        System.out.println("JDBC: MYSQL_PASSWORD from env / .env");
        System.out.println("JPA:  spring.datasource.password: ${MYSQL_PASSWORD}");
        System.out.println("git:  .env.example YES, .env NO");
    }
}
```

Expected Output:

```
JDBC: MYSQL_PASSWORD from env / .env
JPA:  spring.datasource.password: ${MYSQL_PASSWORD}
git:  .env.example YES, .env NO
```

---

2. Entity — `@Entity` / `@Id` / `@GeneratedValue`

Definition: An **entity** is a **plain Java class** annotated `@Entity`. `@Table(name = "campus_products")` names the MySQL table. `@Id` is the primary key. `@GeneratedValue(strategy = GenerationType.IDENTITY)` matches MySQL `AUTO_INCREMENT` (you already used that in MYSQL / JDBC).

Purpose: Without `@Entity`, Hibernate has nothing to map. Without `@Id`, JPA cannot tell rows apart. Without IDENTITY, you would assign ids in Java like module 02's `nextId++` — the **database** should own the catalog id.

Copy-paste valid (same package as `@SpringBootApplication` — Maven tree):

```
package in.ksr.campuscatalog;

import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.Table;

@Entity
@Table(name = "campus_products")
public class Product {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;

    @Column(name = "price_inr")
    private int priceInr;
```

```
    public Product() {
    }

    public Product(String name, int priceInr) {
        this.name = name;
        this.priceInr = priceInr;
    }

    public Long getId() {
        return id;
    }

    public void setId(Long id) {
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public int getPriceInr() {
        return priceInr;
    }

    public void setPriceInr(int priceInr) {
        this.priceInr = priceInr;
    }
}
```

No-arg constructor is required (JPA and Jackson). **Getters / setters** stay explicit — no Lombok on the core path. **`Long id`** is null until `save` — then MySQL fills it.

This is the **same snack idea** as module 02's `CanteenItem` and the FS bridge `products` table. Lab kit uses **`campus_products`** so your MYSQL `products` homework stays untouched.

Note:
- Wrong: `import javax.persistence.Entity` — Boot 3 is **`jakarta.persistence`**.
- Wrong: primitive `long id` with IDENTITY — `0` looks like a real id. Use **`Long`** so "not saved yet" is `null`.
- `@Column` is optional when the name already matches; **`price_inr`** is explicit so the SQL log is readable.

Common Error: Forgetting the no-arg constructor after you add an all-args constructor — Hibernate cannot instantiate the row.

Remember: `@Entity` is a **row type**, not a Spring bean by itself. The **repository** is the Spring bean.

Interview: `@GeneratedValue` IDENTITY vs SEQUENCE? IDENTITY = MySQL auto-increment. SEQUENCE is common on Oracle / PostgreSQL. This course uses IDENTITY.

Print the entity annotation checklist: `@Entity`, `@Table(name = "campus_products")`, `@Id`, and `@GeneratedValue IDENTITY`.

```
public class EntityAnnotationPrint {
    public static void main(String[] args) {
        System.out.println("@Entity");
        System.out.println("@Table(name = \"campus_products\")");
        System.out.println("@Id");
        System.out.println("@GeneratedValue IDENTITY");
    }
}
```

Expected Output:

```
@Entity
@Table(name = "campus_products")
@Id
@GeneratedValue IDENTITY
```

Print Boot 3 `jakarta.persistence.Entity` versus old `javax.persistence.Entity`, and that `javax` will not compile on this pom.

```
public class JakartaNotJavax {
    public static void main(String[] args) {
        System.out.println("Boot 3: jakarta.persistence.Entity");
        System.out.println("Old:    javax.persistence.Entity");
        System.out.println("javax will not compile on this pom");
    }
}
```

Expected Output:

```
Boot 3: jakarta.persistence.Entity
Old:    javax.persistence.Entity
javax will not compile on this pom
```

---

3. `JpaRepository` — save / findAll / findById / deleteById

Definition: `JpaRepository<Product, Long>` is a Spring Data **interface**. The first type is the entity. The second is the **id type**. Boot creates the implementation at startup.

Purpose: Without a repository, you would write JDBC again (or an `EntityManager` by hand). Four methods cover the catalog: **create / list / get / delete**.

Copy-paste valid:

```
package in.ksr.campuscatalog;

import org.springframework.data.jpa.repository.JpaRepository;

public interface ProductRepository extends JpaRepository<Product, Long> {
}
```

Empty body is **correct**. You inherit:

Method	SQL idea (Hibernate writes it)
`save(product)`	INSERT (id null) or UPDATE (id set)
`findAll()`	SELECT all rows
`findById(id)`	SELECT by PK → `Optional`
`deleteById(id)`	DELETE by PK

`findById` returns **`Optional<Product>`**. Module 02 looped an `ArrayList`. Here:

```
Product found = products.findById(id).orElse(null);
if (found == null) {
    // 404 — same status idea as module 02
}
```

You do **not** write `findByName` query methods this module (**Later —** if the batch needs search).

Note:
- Wrong: `JpaRepository<Product, Integer>` while `id` is `Long` — id types must match.
- Wrong: calling `save` and expecting a row without a running MySQL — Boot fails at start if the DataSource is wrong.
- Prefer: constructor injection of `ProductRepository` into the controller (next topic). No `new ProductRepository()`.

Common Error: `products.get(0)` like an `ArrayList`. A repository is not a `List`. Use `findAll()` / `findById`.

Remember: `save` on a **new** entity (id `null`) INSERTs. `save` on an entity **with** an id UPDATEs that row.

Interview: Why an interface? Spring Data builds a proxy. You declare *what*; Boot supplies *how*.

Print the four repository methods you call: `save`, `findAll`, `findById`, and `deleteById`.

```
public class RepoFourMethods {
    public static void main(String[] args) {
        System.out.println("save(product)");
        System.out.println("findAll()");
        System.out.println("findById(id)");
        System.out.println("deleteById(id)");
    }
}
```

Expected Output:

```
save(product)
findAll()
findById(id)
deleteById(id)
```

Print `JpaRepository<Product, Long>` and what each type argument means: entity class and `@Id` type.

```
public class RepoGenericsPrint {
    public static void main(String[] args) {
        System.out.println("JpaRepository<Product, Long>");
        System.out.println("Product = entity class");
        System.out.println("Long    = type of @Id");
    }
}
```

Expected Output:

```
JpaRepository<Product, Long>
Product = entity class
Long    = type of @Id
```

---

4. Maven — `starter-data-jpa` + `mysql-connector-j`

Definition: `spring-boot-starter-data-jpa` pulls Hibernate, Spring Data, and `starter-jdbc` (which brings **HikariCP**). `mysql-connector-j` is the MySQL 8 driver (same family as JDBC week's JAR). Keep `spring-boot-starter-web`.

Purpose: Without the JPA starter, `@Entity` / `JpaRepository` will not compile. Without the driver, Boot starts then fails to talk to MySQL. Maven versions stay on the Boot **3.x** parent (course pin **3.5.16**, JDK **17**).

```
<dependencies>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-data-jpa</artifactId>
    </dependency>
    <dependency>
        <groupId>com.mysql</groupId>
        <artifactId>mysql-connector-j</artifactId>
        <scope>runtime</scope>
    </dependency>
</dependencies>
```

Do **not** add `starter-jdbc` yourself — JPA already pulls it. Do **not** add Security, Thymeleaf, or Lombok to pass.

Optional lite validation (topic 7) adds `spring-boot-starter-validation` — **not** required for the kit Done when.

Note:
- Wrong: start.spring.io **4.x** parent — stay on **3.5.x** + Java 17 (Tested Toolchain).
- Wrong: `mysql-connector-java` old artifact name — Boot 3 uses `mysql-connector-j`.
- Prefer: `<scope>runtime</scope>` on the driver (you do not import driver classes).

Remember: Parent POM still pins versions. You rarely type a Hibernate version.

Print the three Maven artifacts this week: `spring-boot-starter-web`, `spring-boot-starter-data-jpa`, and `mysql-connector-j`.

```
public class JpaPomArtifacts {
    public static void main(String[] args) {
        System.out.println("spring-boot-starter-web");
        System.out.println("spring-boot-starter-data-jpa");
        System.out.println("mysql-connector-j");
    }
}
```

Expected Output:

```
spring-boot-starter-web
spring-boot-starter-data-jpa
mysql-connector-j
```

Print what the JPA starter pulls: Hibernate, Spring Data JPA, and HikariCP via `starter-jdbc`.

```
public class JpaStarterPulls {
    public static void main(String[] args) {
        System.out.println("Hibernate (JPA provider)");
        System.out.println("Spring Data JPA");
        System.out.println("HikariCP via starter-jdbc");
    }
}
```

Expected Output:

```
Hibernate (JPA provider)
Spring Data JPA
HikariCP via starter-jdbc
```

---

5. `application.yml` — datasource, env, Hikari, lab `ddl-auto`

Definition: **`spring.datasource.url`** / **`username`** / **`password`** tell Boot how to reach MySQL. Placeholders **`${MYSQL_PASSWORD}`** read the **environment** (same secret habit as JDBC). **HikariCP** is the connection **pool** Boot auto-configures — a small set of reusable JDBC connections.

Definition: **`spring.jpa.hibernate.ddl-auto`** controls schema from entities. **`update`** means: create missing tables / columns to match `@Entity`. That is **lab-only**. Production schema (Flyway / Liquibase / DBA DDL) is **Later —**.

Purpose: Hard-coding `root` / `root123` in yml and pushing to GitHub is the JDBC-week failure mode again. Env keeps the password off GitHub. Naming Hikari answers the interview; **do not** set `maximum-pool-size` this module.

```
spring:
  application:
    name: campus-jpa-catalog
  datasource:
    url: jdbc:mysql://${MYSQL_HOST:localhost}:${MYSQL_PORT:3306}/${MYSQL_DATABASE:training_db}?allowPub
    username: ${MYSQL_USER:root}
    password: ${MYSQL_PASSWORD}
  jpa:
    hibernate:
      ddl-auto: update
```

```
    show-sql: true
    properties:
      hibernate:
        format_sql: true

server:
  port: ${SERVER_PORT:8080}
```

`allowPublicKeyRetrieval=true` and `useSSL=false` are the **same lab URL flags** as JDBC CRUD. `useSSL=false` is **lab-only**.

`.env.example` (commit this):

```
# Copy keys into your shell or IDE env. Boot does not load this file by itself.
# Do not commit a real .env.
SERVER_PORT=8080
MYSQL_HOST=localhost
MYSQL_PORT=3306
MYSQL_USER=root
MYSQL_PASSWORD=change_me
MYSQL_DATABASE=training_db
```

Export then run (macOS / Linux):

```
export MYSQL_PASSWORD='your_lab_password'
export MYSQL_USER=root
mvn spring-boot:run
```

Windows PowerShell: `$env:MYSQL_PASSWORD='your_lab_password'` then `mvn spring-boot:run`.

Boot **does not** read `.env` automatically — export or IDE Run Configuration env. JDBC kit could parse `.env` in `main`; Boot does not.

Note:
- Wrong: `password: root123` in yml on GitHub.
- Wrong: `ddl-auto: create-drop` for a catalog you care about — **drop** wipes tables on shutdown.
- Wrong: `ddl-auto: create` on every restart in a shared `training_db` — it can smash tables. **`update` is the lab setting.**
- Prefer: `show-sql: true` in class so you **see** Hibernate's INSERT next to JDBC memories.

Common Error: `Public Key Retrieval is not allowed` — same JDBC lab fix on the URL. Do not paste the password into git while debugging.

Remember: Hikari = pool. You name it. You do **not** tune it here.

Interview: Why a pool? Opening a TCP connection per HTTP request is slow. Hikari reuses connections. Tuning is **Later —** / ops.

Print the datasource URL, username, password `${MYSQL_PASSWORD}`, and `spring.jpa.hibernate.ddl-auto=update` keys.

```
public class DatasourceYmlKeys {
    public static void main(String[] args) {
        System.out.println("spring.datasource.url");
```

```
        System.out.println("spring.datasource.username");
        System.out.println("spring.datasource.password=${MYSQL_PASSWORD}");
        System.out.println("spring.jpa.hibernate.ddl-auto=update");
    }
}
```

Expected Output:

```
spring.datasource.url
spring.datasource.username
spring.datasource.password=${MYSQL_PASSWORD}
spring.jpa.hibernate.ddl-auto=update
```

Print that lab uses `ddl-auto=update`, `create-drop` wipes data, and production schema stays Later.

```
public class DdlAutoLabOnly {
    public static void main(String[] args) {
        System.out.println("Lab: ddl-auto=update");
        System.out.println("Not lab: create-drop wipes data");
        System.out.println("Production schema: Later — not this module");
    }
}
```

Expected Output:

```
Lab: ddl-auto=update
Not lab: create-drop wipes data
Production schema: Later — not this module
```

---

6. REST against the DB — same mappings, `JpaRepository` store

Definition: The controller from module 02 still uses **`@GetMapping`**, **`@PostMapping`**, **`@PathVariable`**, **`@RequestBody`**, **200 / 201 / 404**. The field is no longer `List<CanteenItem>`. It is a **`ProductRepository`**. Spring **constructor-injects** it (one constructor — no `@Autowired` required).

Purpose: Learners already know HTTP. Today GET list is `findAll()`. GET by id is `findById`. POST is `save` with id `null` so MySQL IDENTITY assigns the key.

Copy-paste valid:

```
package in.ksr.campuscatalog;

import java.util.List;
import java.util.Map;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.ResponseStatus;
import org.springframework.web.bind.annotation.RestController;

@RestController
```

```java
@RequestMapping("/products")
public class ProductController {

    private final ProductRepository products;

    public ProductController(ProductRepository products) {
        this.products = products;
    }

    @GetMapping
    public List<Product> list() {
        return products.findAll();
    }

    @GetMapping("/{id}")
    public ResponseEntity<?> getById(@PathVariable Long id) {
        Product found = products.findById(id).orElse(null);
        if (found != null) {
            return ResponseEntity.ok(found);
        }
        return ResponseEntity.status(HttpStatus.NOT_FOUND)
                .body(Map.of("error", "No product " + id));
    }

    @PostMapping
    @ResponseStatus(HttpStatus.CREATED)
    public Product add(@RequestBody Product incoming) {
        incoming.setId(null);
        return products.save(incoming);
    }
}
```

`incoming.setId(null)` stops a client from forcing an UPDATE by sending `"id": 1`. POST **creates**.

`deleteById` is in-window on the **repository**. A **DELETE** HTTP mapping is optional Stretch — core ticket is list / get / create.

Prove persistence:

```
curl http://localhost:8080/products
curl -X POST http://localhost:8080/products \
  -H "Content-Type: application/json" \
  -d '{"name":"Pani puri","priceInr":30}'
# Ctrl+C the server, mvn spring-boot:run again, then:
curl http://localhost:8080/products
```

Pani puri must still be there. That is the whole point.

Note:
- Wrong: keeping `new ArrayList<>()` in the controller "and also" a repository — two stores, confusing GET.
- Wrong: returning `products.toString()` — Jackson needs the `List<Product>`.
- Paths here are `/products`. Module 02 used `/canteen/items`. Same HTTP ideas; CampusCart catalog name.

Common Error: POST without `setId(null)` and a body that includes `"id": 5` — Hibernate may try to UPDATE a missing row. Clear the id on create.

Remember: Constructor injection = Spring passes the repository. You never `new` it.

Interview: `Optional` from `findById`? Empty means no row. Map that to **404**, not `null` with **200**.

Print HTTP-to-repository mapping: GET list `findAll`, GET by id `findById`, POST `save` with null id and 201.

```
public class HttpToRepoPrint {
    public static void main(String[] args) {
        System.out.println("GET  /products      -> findAll");
        System.out.println("GET  /products/{id} -> findById");
        System.out.println("POST /products      -> save (id null) -> 201");
    }
}
```

Expected Output:

```
GET  /products      -> findAll
GET  /products/{id} -> findById
POST /products      -> save (id null) -> 201
```

Print that POST Pani puri lands in MySQL `campus_products`, a Tomcat restart still shows it, and that failed with ArrayList in module 02.

```
public class RestartKeepsRow {
    public static void main(String[] args) {
        System.out.println("POST Pani puri -> MySQL campus_products");
        System.out.println("Restart Tomcat -> GET still shows Pani puri");
        System.out.println("That failed with ArrayList in module 02");
    }
}
```

Expected Output:

```
POST Pani puri -> MySQL campus_products
Restart Tomcat -> GET still shows Pani puri
That failed with ArrayList in module 02
```

---

7. Lite validation on the entity (short — not module 04)

Definition: Bean Validation annotations on fields (`@NotBlank`, `@NotNull`) describe **what is allowed**. **`@Valid`** on **`@RequestBody`** tells Spring to **run** those checks before the method body.

Purpose: POST `{"name":"","priceInr":30}` should not become a catalog row. A **one-line** guard on the entity is enough to *name* `@Valid`. **DTO mapping**, **Postman collections**, and **error JSON bodies** are **Later —** module 04 (calendar homework for `@Valid` on POST sits there).

Optional dependency (Stretch / peek — **not** kit Done when):

```
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-validation</artifactId>
</dependency>
```

On the entity (imports `jakarta.validation.constraints`):

```
@NotBlank
private String name;

@NotNull
private Integer priceInr;
```

Use **`Integer`** (wrapper) if you want `@NotNull` — a primitive `int` is never null, so `@NotNull` would not catch a missing JSON field the way you think.

On POST (peek):

```
@PostMapping
@ResponseStatus(HttpStatus.CREATED)
public Product add(@Valid @RequestBody Product incoming) {
    incoming.setId(null);
    return products.save(incoming);
}
```

Default 400 body from Boot is **ugly JSON**. Designing that body is **Later —** module 04. Core path may skip `@Valid` entirely.

Note:
- Wrong: treating this peek as a DTO layer — the JSON **is** still the entity.
- Wrong: `@Valid` without `starter-validation` — annotations will not run (or will not compile).
- Prefer: kit **Done when** without `@Valid`; add it only as Stretch.

Remember: Filename includes Validation. Depth of `@Valid` + error bodies is the **next** module.

Print the lite checks: `@NotBlank` on name, `@NotNull` on the price wrapper, and `@Valid` on the POST body.

```
public class LiteValidPrint {
    public static void main(String[] args) {
        System.out.println("@NotBlank on name");
        System.out.println("@NotNull on price wrapper");
        System.out.println("@Valid on POST body");
    }
}
```

Expected Output:

```
@NotBlank on name
@NotNull on price wrapper
@Valid on POST body
```

Print what stays Later: DTO mapping, Postman collection, and a designed error JSON body.

```
public class ValidLaterPrint {
    public static void main(String[] args) {
        System.out.println("Later: DTO mapping");
        System.out.println("Later: Postman collection");
        System.out.println("Later: designed error JSON body");
```

```
    }
}
```

Expected Output:

```
Later: DTO mapping
Later: Postman collection
Later: designed error JSON body
```

Step-by-Step Annotated Java Code

Work in lab kit **18_jpa_product_crud** (`starter/`). JDK 17. Open the folder that contains **`pom.xml`**. MySQL 8+ must be running. Copy `.env.example` values into your **shell or IDE env** (Boot does not load `.env` by itself).

Step 1: Prove MySQL and the pom

Confirm MySQL (same habit as JDBC week):

```
mysql -u root -p -e "SHOW DATABASES;"
```

Create `training_db` if needed (`CREATE DATABASE IF NOT EXISTS training_db;` — MYSQL/00). Optional: run kit **`seed/campus_products.sql`** after the first Boot start, or let **`ddl-auto=update`** create `campus_products` from the entity.

Open `starter/pom.xml`. You need **`starter-web`**, **`starter-data-jpa`**, and **`mysql-connector-j`**.

Print database `training_db`, table `campus_products`, and that you must not drop the MYSQL products homework table.

```
public class TrainingDbName {
    public static void main(String[] args) {
        System.out.println("training_db");
        System.out.println("table: campus_products");
        System.out.println("do not drop MYSQL products homework");
    }
}
```

Expected Output:

```
training_db
table: campus_products
do not drop MYSQL products homework
```

Print the three dependencies again: `starter-web`, `starter-data-jpa`, and `mysql-connector-j`.

```
public class StarterPomCheck {
    public static void main(String[] args) {
        System.out.println("starter-web");
        System.out.println("starter-data-jpa");
        System.out.println("mysql-connector-j");
    }
}
```

Expected Output:

```
starter-web
starter-data-jpa
mysql-connector-j
```

Step 2: `application.yml` + env (no password in git)

Complete datasource keys. `password: ${MYSQL_PASSWORD}` with **no** literal. `.gitignore` already has `.env` and `target/` (Git kit).

```
export MYSQL_PASSWORD='your_lab_password'
cd starter
mvn spring-boot:run
```

If MySQL is down or the password is wrong, Boot **fails at start** — fix env, do not paste the password into yml.

Print `export MYSQL_PASSWORD=...`, then `mvn spring-boot:run`, and that Boot does not read `.env` by itself.

```
public class ExportMysqlPassword {
    public static void main(String[] args) {
        System.out.println("export MYSQL_PASSWORD=...");
        System.out.println("mvn spring-boot:run");
        System.out.println("Boot does not read .env by itself");
    }
}
```

Expected Output:

```
export MYSQL_PASSWORD=...
mvn spring-boot:run
Boot does not read .env by itself
```

Print that the pool name is HikariCP, Boot auto-configures it, and you do not set `maximum-pool-size` this module.

```
public class HikariNameOnly {
    public static void main(String[] args) {
        System.out.println("Pool name: HikariCP");
        System.out.println("Auto-configured by Boot");
        System.out.println("Do not set maximum-pool-size this module");
    }
}
```

Expected Output:

```
Pool name: HikariCP
Auto-configured by Boot
Do not set maximum-pool-size this module
```

Step 3: `Product` entity

Annotate `Product` with `@Entity`, `@Table(name = "campus_products")`, `@Id`, `@GeneratedValue(IDENTITY)`, `@Column(name = "price_inr")`. No-arg constructor + getters / setters. **`jakarta.persistence`**.

Print the Product fields: `id` Long IDENTITY, `name` String, and `priceInr` mapped to column `price_inr`.

```
public class ProductFieldsPrint {
    public static void main(String[] args) {
        System.out.println("id Long IDENTITY");
        System.out.println("name String");
        System.out.println("priceInr -> column price_inr");
    }
}
```

Expected Output:

```
id Long IDENTITY
name String
priceInr -> column price_inr
```

Print why you need a no-arg constructor (JPA + Jackson), a name+price constructor for demos, and that MySQL assigns `id` on save.

```
public class EntityConstructorsPrint {
    public static void main(String[] args) {
        System.out.println("no-arg: JPA + Jackson");
        System.out.println("name + priceInr: handy for demos");
        System.out.println("id assigned by MySQL on save");
    }
}
```

Expected Output:

```
no-arg: JPA + Jackson
name + priceInr: handy for demos
id assigned by MySQL on save
```

Step 4: `ProductRepository`

Empty interface `extends JpaRepository<Product, Long>`. Same package as the entity so component scan finds it (`@SpringBootApplication` already scans `in.ksr.campuscatalog`).

Print that `ProductRepository` extending `JpaRepository<Product, Long>` with an empty body is enough — no method bodies required.

```
public class EmptyRepoOk {
    public static void main(String[] args) {
        System.out.println("public interface ProductRepository");
        System.out.println("extends JpaRepository<Product, Long> { }");
        System.out.println("no method body required");
    }
}
```

Expected Output:

```
public interface ProductRepository
extends JpaRepository<Product, Long> { }
no method body required
```

Print that `CampusCatalog` application, entity, repo, and controller share `in.ksr.campuscatalog`, and a wrong package means bean not found.

```
public class SamePackageScan {
    public static void main(String[] args) {
        System.out.println("Application: in.ksr.campuscatalog");
        System.out.println("Entity + repo + controller: same package");
        System.out.println("Wrong package -> bean not found");
    }
}
```

Expected Output:

```
Application: in.ksr.campuscatalog
Entity + repo + controller: same package
Wrong package -> bean not found
```

Step 5: Controller + curl (list / get / create)

Replace any leftover `ArrayList` with constructor injection of `ProductRepository`. Complete GET list, GET by id (404), POST 201. Then:

```
curl http://localhost:8080/products
curl http://localhost:8080/products/1
curl http://localhost:8080/products/99
curl -X POST http://localhost:8080/products \
  -H "Content-Type: application/json" \
  -d '{"name":"Pani puri","priceInr":30}'
```

Watch the terminal: `show-sql: true` prints Hibernate SQL. That is your JDBC week in the log.

Print curl for `GET http://localhost:8080/products` and that the body is a JSON array from `findAll`.

```
public class CurlListPrint {
    public static void main(String[] args) {
        System.out.println("curl http://localhost:8080/products");
        System.out.println("JSON array from findAll");
    }
}
```

Expected Output:

```
curl http://localhost:8080/products
JSON array from findAll
```

Print that POST JSON `name` plus `priceInr` returns 201 and an id from `AUTO_INCREMENT`.

```
public class CurlPostPrint {
    public static void main(String[] args) {
        System.out.println("POST JSON name + priceInr");
        System.out.println("201 and id from AUTO_INCREMENT");
    }
}
```

Expected Output:

```
POST JSON name + priceInr
201 and id from AUTO_INCREMENT
```

Step 6: Restart proof + git hygiene

Stop the app. Start it again. GET `/products` must still include Pani puri. `git status` must **not** list `.env` or `target/`. README: JDK 17, Boot 3, env vars, `mvn spring-boot:run`, curl.

Do **not** add Security, DTOs, Postman, or Lombok to pass. `@Valid` is Stretch only.

Print what to ignore (`target/`, `.env`) and what to keep (`pom.xml`, `src/`, `.env.example`).

```
public class JpaGitignorePrint {
    public static void main(String[] args) {
        System.out.println("ignore: target/");
        System.out.println("ignore: .env");
        System.out.println("keep: pom.xml src/ .env.example");
    }
}
```

Expected Output:

```
ignore: target/
ignore: .env
keep: pom.xml src/ .env.example
```

Print the core ticket: `Product` entity plus `ProductRepository`, GET list / GET by id / POST create, stored in MySQL not an ArrayList.

```
public class Kit18CoreTicket {
    public static void main(String[] args) {
        System.out.println("Product entity + ProductRepository");
        System.out.println("GET list / GET by id / POST create");
        System.out.println("MySQL — not ArrayList");
    }
}
```

Expected Output:

```
Product entity + ProductRepository
GET list / GET by id / POST create
MySQL — not ArrayList
```

Practice Exercises

(Each Practice item includes a filled Model answer: and Expected Output: where the demo prints.)

Basic

1. Print JDBC vs JPA in two lines.

Model answer:

```
public class JdbcJpaTwoLines {
    public static void main(String[] args) {
        System.out.println("JDBC: you write PreparedStatement SQL");
        System.out.println("JPA:  Hibernate writes SQL from @Entity");
    }
}
```

Expected Output:

```
JDBC: you write PreparedStatement SQL
JPA:  Hibernate writes SQL from @Entity
```

2. Print the three entity annotations that must appear on `Product`.

Model answer:

```
public class ThreeEntityAnnotations {
    public static void main(String[] args) {
        System.out.println("@Entity");
        System.out.println("@Id");
        System.out.println("@GeneratedValue");
    }
}
```

Expected Output:

```
@Entity
@Id
@GeneratedValue
```

3. Print the repository type line.

Model answer:

```
public class RepoTypeLine {
    public static void main(String[] args) {
        System.out.println("interface ProductRepository extends JpaRepository<Product, Long>");
    }
}
```

Expected Output:

```
interface ProductRepository extends JpaRepository<Product, Long>
```

Intermediate

1. Sketch yml password: placeholder vs literal.

Model answer:

```
public class PasswordPlaceholderSketch {
    public static void main(String[] args) {
        System.out.println("OK:   password: ${MYSQL_PASSWORD}");
        System.out.println("NO:   password: root123");
    }
}
```

Expected Output:

```
OK:   password: ${MYSQL_PASSWORD}
NO:   password: root123
```

2. Sketch GET by id with JPA: method, miss path.

Model answer:

```
public class JpaGetByIdSketch {
    public static void main(String[] args) {
        System.out.println("findById(id).orElse(null)");
        System.out.println("null -> 404");
        System.out.println("found -> 200 JSON");
    }
}
```

Expected Output:

```
findById(id).orElse(null)
null -> 404
found -> 200 JSON
```

3. Why `setId(null)` on POST?

Model answer:

```
public class PostClearId {
    public static void main(String[] args) {
        System.out.println("POST must INSERT");
        System.out.println("null id -> IDENTITY");
        System.out.println("client id would look like UPDATE");
    }
}
```

Expected Output:

```
POST must INSERT
null id -> IDENTITY
client id would look like UPDATE
```

4. Why is `ddl-auto=update` labelled lab-only?

Model answer:

```
public class WhyUpdateLabOnly {
    public static void main(String[] args) {
        System.out.println("update changes tables from Java");
        System.out.println("production needs a planned schema");
        System.out.println("Later — migrations / DBA DDL");
    }
}
```

Expected Output:

```
update changes tables from Java
production needs a planned schema
Later — migrations / DBA DDL
```

Interview-Oriented

1. JDBC vs JPA — one sentence each?

Model answer: JDBC is explicit SQL through `PreparedStatement`. JPA maps entities; Hibernate generates SQL and still uses JDBC (plus a pool).

2. Why `jakarta.persistence`?

Model answer: Spring Boot 3 moved from Java EE `javax.*` to Jakarta. Entity annotations live in `jakarta.persistence`. `javax.persistence` will not compile on this parent.

3. What is HikariCP here?

Model answer: The JDBC connection pool Boot auto-configures when a DataSource is present. Name it in interviews. Do not tune pool size in this module.

4. `save` vs `findAll` vs `findById`?

Model answer: `save` persists (INSERT or UPDATE). `findAll` returns every row as a `List`. `findById` returns `Optional` for one primary key.

5. Does JPA mean we skip MYSQL week?

Model answer: No. You still design tables, PKs, and joins in SQL. JPA is how this Boot API talks to those tables. Point at `MYSQL/` — do not rewrite it.

6. Is `@Valid` required to pass this assignment?

Model answer: No. Lite `@NotBlank` / `@Valid` is a short peek. Kit Done when is entity + repository + list / get / create on MySQL. Error bodies and DTOs are **Later —** module 04.

Mini-Project (Assignment): Swiggy / CampusCart JPA Product Catalog

Lab kit: 15_Lab_Assets/18_jpa_product_crud/

Module 02 proved HTTP on an `ArrayList`. This assignment is the **portfolio JPA spike**: a **`Product` entity**, a **`JpaRepository`**, and **GET list / GET by id / POST** that **survive restart** in MySQL. Push a public repo. No secrets in git.

Do **not** add Security, DTO classes, Postman collections, or Lombok to pass. `@Valid` is Stretch only.

What to build

Piece	Detail
`pom.xml`	Parent **Spring Boot 3.x**, `<java.version>17</java.version>`, **`starter-web`**, **`starter-data-jpa`**, **`mysql-connector-j`**
Application class	`@SpringBootApplication` + `SpringApplication.run`
`Product`	`@Entity`, `@Table(name = "campus_products")`, `@Id`, `@GeneratedValue(IDENTITY)`, `name`, `priceInr` — explicit getters / setters, no-arg constructor, **`jakarta.persistence`**
`ProductRepository`	`extends JpaRepository<Product, Long>`
GET list	**GET** **`/products`** → JSON **array** from **`findAll()`**
GET by id	**GET** **`/products/{id}`** — `@PathVariable`; **200** or **404** via **`findById`**
POST	**POST** **`/products`** — `@RequestBody`; **201**; **`save`** after `setId(null)`
`application.yml`	`spring.datasource.*` from **`${MYSQL_*}`**; **`ddl-auto: update`**; `show-sql` OK
`.env.example`	Dummy `MYSQL_PASSWORD` / host / database — **no** real password
`.gitignore`	`target/`, `.env`, `*.class`, `.idea/`
README	JDK 17, Boot 3, export env, `mvn spring-boot:run`, curl list / by-id / POST, restart proof

SQL depth stays in `MYSQL/`. Kit **`seed/campus_products.sql`** is optional insert help after the table exists — not a second SQL course.

Sample curl (after at least one POST):

```
[{"id":1,"name":"Idli","priceInr":40},{"id":2,"name":"Filter coffee","priceInr":25}]
```

```
{"id":1,"name":"Idli","priceInr":40}
```

```
{"id":3,"name":"Pani puri","priceInr":30}
```

Work from `starter/` (TODOs) or `broken/` (debug). Compare `solution/` only after you finish (or when told in class). `solution/` is trainer-only in the student pack.

Stretch

- Lite `@Valid` + `@NotBlank` / `@NotNull` (`starter-validation`) — **not** Done when.
- Extra field (e.g. `category`) on entity + column — **not** Done when.
- HTTP **DELETE** `/products/{id}` calling `deleteById` — optional.
- Lombok getters — optional, not in Done when.
- **Later —** DTOs, Postman collection, error bodies (module 04). **Later —** Security.

Done when

- [ ] `mvn spring-boot:run` starts (Tomcat + DataSource; MySQL reachable)
- [ ] `Product` is an `@Entity` with `@Id` + `@GeneratedValue` (`jakarta.persistence`)
- [ ] `ProductRepository` extends `JpaRepository<Product, Long>`
- [ ] GET `/products` returns a JSON **array** from the **database** (`findAll`)
- [ ] GET `/products/{id}` uses `@PathVariable` + `findById` — 200 or **404**
- [ ] POST `/products` uses `@RequestBody` + `save` and returns **201**
- [ ] Restart the app; POST data is **still** in GET list
- [ ] Password is **not** a literal in yml or source; `.env` not committed
- [ ] No Spring Security / JWT, no DTO layer, no Postman collection required
- [ ] No Lombok required to pass; `@Valid` not required to pass
- [ ] `target/` untracked; short README with run + curl (Standard Project Deliverables)

Interview Q&A

Shared with students after concepts and the mini-project. Use for review and interview prep — questions and model answers are included.

Q1. What is JPA?
A: Jakarta Persistence API — the standard for mapping Java classes to relational tables (`@Entity`, `@Id`). Hibernate is the implementation Boot uses.

Q2. JDBC vs JPA?
A: JDBC: you write SQL with `PreparedStatement`. JPA: you map objects; Hibernate generates SQL and still uses JDBC. This course taught JDBC first on purpose.

Q3. What is an entity?
A: A class annotated `@Entity` whose instances correspond to rows. Fields map to columns. It needs a primary key (`@Id`).

Q4. Why `@GeneratedValue(IDENTITY)`?
A: It matches MySQL `AUTO_INCREMENT`. The database assigns the id on INSERT. Leave id `null` on create.

Q5. `javax.persistence` vs `jakarta.persistence`?
A: Boot 3 / Jakarta EE 9+ uses `jakarta.*`. Old tutorials show `javax.persistence` — that does not compile on this pom.

Q6. What is `JpaRepository`?

A: A Spring Data interface. `JpaRepository<Product, Long>` gives `save`, `findAll`, `findById`, `deleteById` without you writing those methods.

Q7. What does `save` do?

A: If the id is null, INSERT. If the id exists, UPDATE that row. POST should null the id so it stays a create.

Q8. What is HikariCP?

A: Boot's default JDBC connection pool (via `starter-jdbc` inside `starter-data-jpa`). It reuses connections. Do not tune pool settings in this module.

Q9. Why `application.yml` for the URL but env for the password?

A: Non-secret defaults (host, port, database name) can live in yml with placeholders. The password is a secret — `${MYSQL_PASSWORD}` from the environment. Commit `.env.example` only.

Q10. Why is `ddl-auto=update` lab-only?

A: It lets Hibernate change tables from entity fields. Production teams version schema separately (migrations / DBA). `create-drop` wipes data. This course uses `update` in class only.

Q11. Does Boot load `.env` automatically?

A: No. It reads environment variables (and yml). Export keys or set them in the IDE. Same story as module 01.

Q12. Why not `DriverManager` in the controller?

A: Boot already configured a DataSource (Hikari). The repository uses it. Mixing raw JDBC in the controller skips that and leaks connections easily.

Q13. In-memory `List` vs JPA?

A: List dies on restart and is one JVM. JPA + MySQL is shared durable catalog. Module 02 used List to learn HTTP first.

Q14. Do we need DTOs this module?

A: No. Returning the entity as JSON is acceptable for this spike. DTO mapping and Postman are **Later —** module 04.

Q15. Is `@Valid` required?

A: Not on the core path. A short entity-level `@NotBlank` / `@Valid` peek is optional. Designed 400 bodies are the next module.

Q16. Will POST survive restart?

A: Yes — that is the acceptance test. Rows live in `campus_products` in `training_db`.

Q17. How does this connect to CampusCart?

A: Capstone catalog (list / detail) is this shape: `Product` entity + REST + MySQL. Auth and orders come later.

Next

Push **CampusCart JPA Product Catalog** (kit 18 — no secrets, `target/` ignored). Continue with **REST, DTOs, and Postman** — DTO mapping, `@Valid` in depth, Postman collection, error bodies. Security stays after that.